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METHEMOGLOBINEMIA DISEASES

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Methemoglobinemia disease may arise from a variety of etiologies including genetic, dietary, idiopathic, and toxicologic sources. Hemoglobin molecules contain iron within a Porphyrinheme structure. The iron in hemoglobin is normally found in the Fe^{2+} state. The iron moiety of hemoglobin can be oxidized to the Fe^{3+} state to form methemoglobinemia. However, this is not the case, because allosteric changes to the hemoglobin molecule cause oxygen to bind more tightly in the partially oxidized hemoglobin molecules. This increased affinity causes a left shift in the hemoglobin-oxygen saturation curve. To combat this process, there are several enzyme systems in the RBC that inhibit oxidation or reduce MHb back to hemoglobin. The cytochrome-b5–MHb reductase system is the predominant system and accounts for approximately 99 % of daily MHb reduction. Ascorbic acid and glutathione account for small amounts of reduction. Under normal circumstances, these enzymes/molecules play a minor role in reduction. However, patients with methemoglobinemia as a result of complete congenital cytochrome-b5 reductase deficiency are able to maintain MHb levels below 50 %.

Oxygen transport, the major function of hemoglobin, is dependent upon reduced heme iron. In the red cell, the heme iron is maintained in the reduced form by the methemoglobin reduction system. When the balance between oxidation and reduction of heme-iron is perturbed due to the presence of excessive oxidants, decreased reducing capacity or the presence of abnormal hemoglobin, methemoglobinemia ensues. In most cases methemoglobinemia is transitory and of no major clinical consequence. Occasionally, however, it can be life threatening and must be rapidly diagnosed and treated. When methemoglobinemia is of hereditary nature, either due to deficiency of red cell NADH-methemoglobin reductase or due to the presence of M hemoglobin, it is a lifelong problem. Since most of these patients do not have major disabling symptoms, the treatment is aimed at correction of cyanosis

Another enzyme that reduces MHb is reduced nicotinamide adenine dinucleotide phosphate (NADPH)–MHb reductase. Also known as NADPH flavinreductase and NADPH-MHb–diaphorase, under normal conditions this enzyme plays a negligible role in reducing MHb. It is actually a generalized reductase with an affinity for dyes, such as methylene blue, Nile blue, and divicine.12-14 In the presence of the cofactor NADPH, this enzyme will reduce these dyes, which in turn reduce MHb. The primary function of this reductase is probably to metabolize oxidant xenobiotics and not MHb. It is somewhat fortuitous that the reduced form of methylene blue has a high affinity for MHb. NADPH-MHb reductase deficiency has also been described, but does not lead to methemoglobinemia.10 This enzyme is not responsible for endogenous MHb reduction, and only reduces MHb in the presence of an exogenous catalyzing agent such as methylene blue. When patients with this disorder develop chemically induced methemoglobin, they do not respond to conventional methylene blue therapy, because methylene blue is dependent on this enzyme to reduce MHb.

Protective mechanisms against oxidative stress include sulfation enzymes, ascorbic acid, and glutathione. These enzymes and peptides serve to detoxify oxidative exogenous chemicals and thereby indirectly prevent methemoglobinemia. Reduced glutathione is quantitatively the most important cellular antioxidant, and is of key importance in all cells for the preservation of protein sulfhydryl groups and to prevent oxidative damage in general. Glutathione is a minor pathway in the reduction of MHb. Glutathione may also metabolize potentially toxic xenobiotics to nontoxic intermediates by forming mercapturic acid conjugates. Oxidized glutathione is cytotoxic and will diffuse out of cells if it is not reduced back to reduced glutathione. Hence, oxidative stress in the presence of glucose-6-phosphate dehydrogenase (G6PD) deficiency leads to intracellular depletion of total glutathione. Other enzymes involved in free radical metabolism include superoxide dismutase, catalase, and glutathione peroxidase. Despite the role these enzymes play in detoxifying agents that cause methemoglobinemia, congenital deficiencies of virtually all these proteins have been described, and none are associated with methemoglobinemia. This is likely due to the extraordinary efficiency of the cytochrome b5/cytochrome-b5 reductase system in reducing MHb.